ONE FOR ALL: CRYO-EM TO CRYO-ET WITH THE NEW GLACIOS 3

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Cryo-electron microscopy (cryo-EM) is an established method for determining protein structures and peering into their function. Recent advances in cryo-EM are making structural determination more accessible and effective. The Glacios 3 combines AI-powered Smart EPU data collection with a 200 kV cold field-emission gun, Selectris post-column energy filter, and Falcon 4i direct electron detector to support high-quality single particle analysis. A new environmental enclosure regulates temperature, vibration, and electromagnetic disturbances, improving imaging stability and enabling installation in a wider range of locations. Additionally, some biological questions cannot be addressed with *in vitro*samples and rely on understanding their biological context. Fo this reason, the Glacios 3 is also designed for cryo-electron tomography (cryo-ET), with a 12-sample autoloader and tomography software for batch multi-sample acquisition and on-the-fly tomogram reconstruction. By combining cryo-EM and cryo-ET capabilities, the Glacios 3 is the most effective tool for both structural and cellular biology.